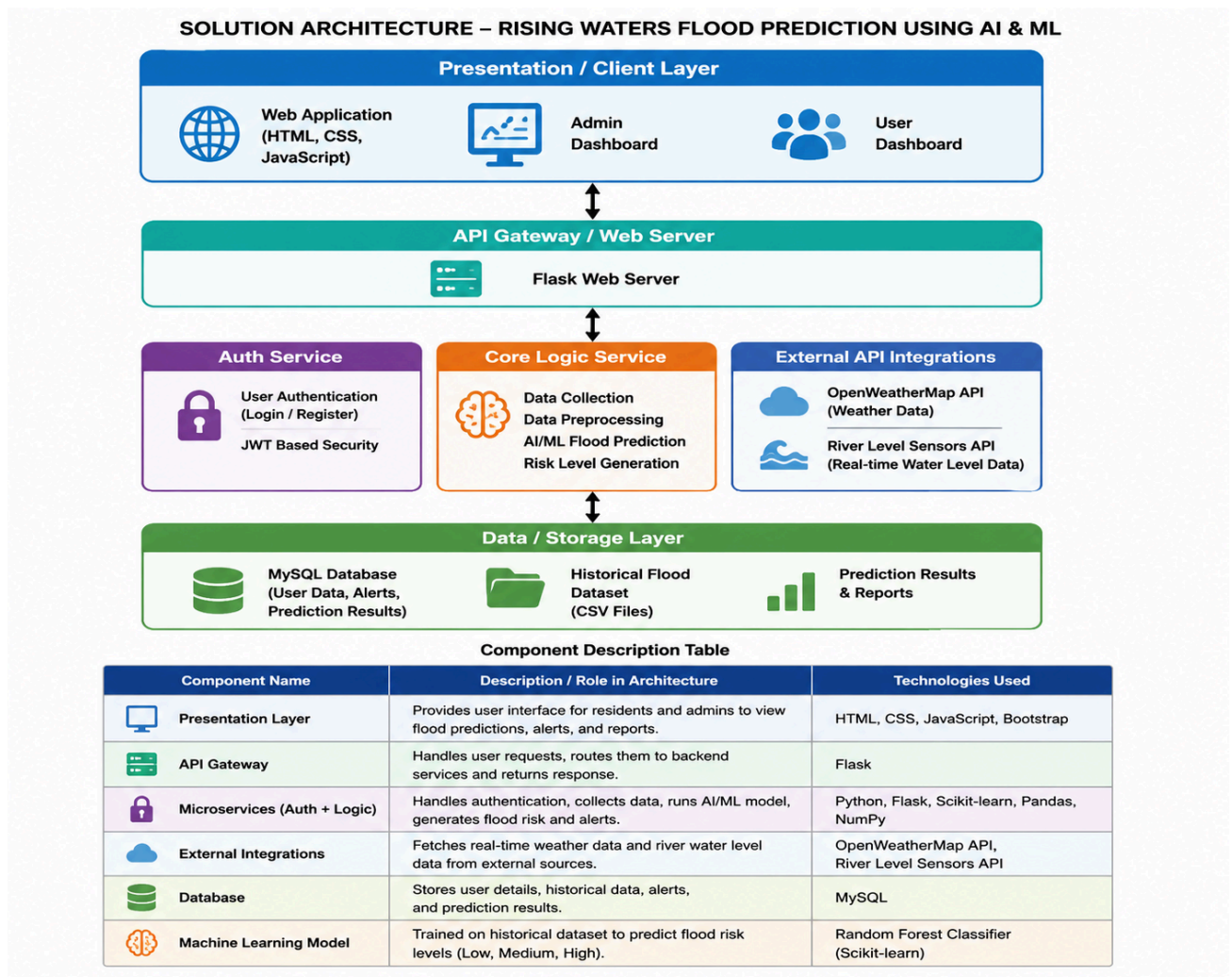


Solution Architecture

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides a web interface where users and administrators can view flood predictions, alerts, and reports.</i>	<i>HTML, CSS, JavaScript, Bootstrap</i>
<i>API Gateway</i>	<i>Receives requests from users, forwards them to the backend, and returns prediction results.</i>	<i>Flask</i>
<i>Application / Core Logic Service</i>	<i>Collects weather and river-level data, preprocesses data, runs the AI/ML model, predicts flood risk, and generates alerts.</i>	<i>Python, Scikit-learn, Pandas, NumPy</i>
<i>[Database]</i>	<i>Stores historical flood data, user information, weather records, and prediction results</i>	<i>MySQL</i>